6 **(a)** **(i)** Write down the equation defining *magnetic flux density* in terms of F the force it produces on a long, straight conductor of length L carrying a current I at an angle θ to the field.

[1]

(ii) Draw a diagram to illustrate the direction of the force relative to the current and magnetic field.

[1]

(b) Fig. 6.1 shows a small square coil of N turns and sides of length L. It is mounted so that it can pivot freely through the centre of the coil, about a horizontal axis PQ parallel to one pair of sides of the coil.

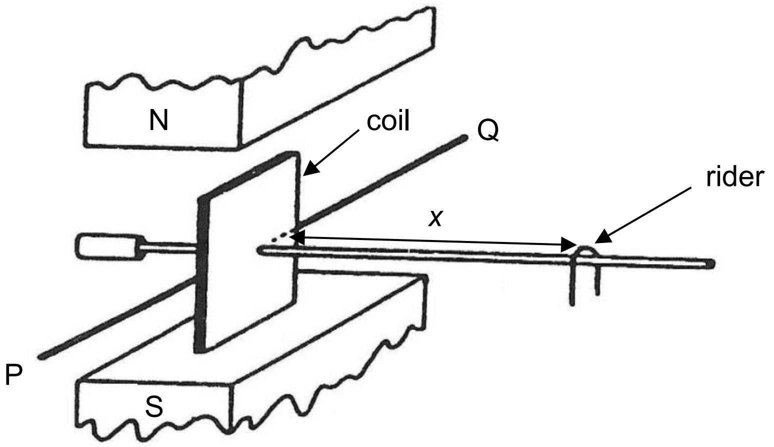

Fig. 6.1

The coil is situated between the poles of a magnet which produces a uniform vertical magnetic field of flux density B. The coil is maintained in a vertical plane by moving a rider of mass M along a horizontal beam attached to the coil. When a current I flow through the coil, equilibrium is restored by adjusting x, the distance of the rider from the coil.

(i) Starting from the definition of magnetic flux density, show that B is given by the expression

$$B = \frac{Mgx}{IL^2N}.$$

[3]

(ii) Current I is supplied by a battery of constant e.m.f. and negligible internal resistance. Discuss the effect on x if the coil is replaced by one wound with wire of same material and diameter, but forming a coil of N turns with sides of length $\frac{L}{2}$.

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.... [3]